Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Project Documentation

1. Introduction

- **Project Title:** Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
- **Team Members:**
  - Team Member 1: Project lead
  - Team Member 2: ServiceNow developer (catalog and flow build)
  - Team Member 3: Business analyst (requirements and test planning)
  - Team Member 4: IT procurement coordinator (process owner and UAT)
  - Team Member 5: Tester (test cases, defect tracking and documentation)

2. Project Overview

- **Purpose:** Automate standard laptop ordering in ServiceNow so employees receive approved devices sooner and IT spends less time on repetitive procurement tasks.
- **Features:**
  - Standard Laptop catalog item with details auto-filled from the user record, plus eligibility and duplicate validation
  - Automatic approval for pre-approved roles, or one-click manager approval with escalation after 48 hours
  - Automatic inventory check and reservation, with a purchase order raised when stock runs out
  - Asset record creation, notifications at every stage and a turnaround and savings dashboard

3. Architecture

- **Presentation layer (in place of Frontend):** Employee Center / Service Portal for placing requests and tracking status, with email and mobile approvals for managers.
- **Application layer (in place of Backend):** ServiceNow Service Catalog and Flow Designer. A main flow calls three subflows: approval, inventory check and purchase order, and asset record and notification.
- **Data layer (in place of Database):** ServiceNow tables: sc_request and sc_req_item for requests, alm_hardware for asset inventory and proc_po for purchase orders. The flows read and update these tables through Flow Designer actions.

4. Setup Instructions

- **Prerequisites:**
  - A ServiceNow instance (for example, a Personal Developer Instance)
  - Administrator access and test users for the employee, manager and coordinator roles
  - Email notifications switched on in the instance, and access to the vendor endpoint for purchase orders
- **Installation:**
  - 1. Request or open your ServiceNow instance and sign in as an administrator.
  - 2. Create the Standard Laptop catalog item with its variables (role, location, department, cost center).
  - 3. Create the roles and assign them to the test users.
  - 4. Build the main flow and the three subflows in Flow Designer, then activate them.
  - 5. Set up the notifications and the vendor REST message with its credentials.
  - 6. Add sample laptop models to the asset inventory and set their stock levels.

5. Folder Structure

ServiceNow keeps work as records inside an application scope rather than in folders, so the table below shows how the components are grouped.

Component	Contents
Catalog (in place of Client)	Standard Laptop catalog item, variables and request form
Flows (in place of Server)	Main laptop order flow and its subflows: approval, inventory check and purchase order, asset record and notification
Actions and rules	Custom actions, decision rules for pre-approved roles and the escalation timer
Notifications	Email templates for approved, shipped and delivered
Reports	Dashboard and reports for turnaround time, request status and savings

6. Running the Application

- **Employee view:** open Employee Center, choose Standard Laptop and submit the request.
- **Flow check:** go to Flow Designer, then Executions, to trace the request step by step.
- **Manager view:** approve the task from the email or from the approvals list.
- **Coordinator view:** review stock, purchase orders and asset records.

7. API Documentation

The solution uses ServiceNow's Table API to read records and one outbound REST message to send purchase orders to the vendor. The endpoints below are examples.

Method	Endpoint	Parameters	Description
GET	`/api/now/table/sc_request`	sysparm_query, sysparm_limit	Retrieves laptop requests, for example filtered by state
GET	`/api/now/table/alm_hardware`	sysparm_query	Looks up stock for a laptop model
POST	`[vendor endpoint]/purchase-orders`	Model, quantity, ship-to location	Submits a purchase order to the vendor (outbound REST message)

Example response for the vendor purchase order call:

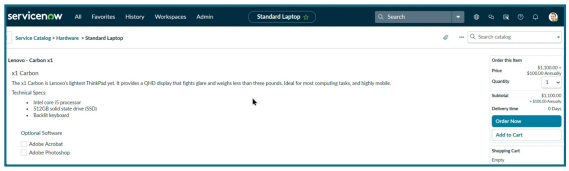
```
{
  "po_number": "PO0004781",
  "status": "confirmed",
  "expected_delivery": "2025-03-19"
}
```

8. Authentication

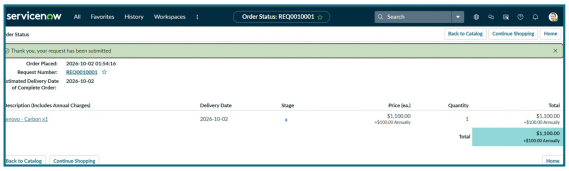
- Users sign in through the identity provider with single sign-on (SAML).
- Authorization relies on ServiceNow roles and access control rules, so only managers can approve and only authorized users can view request and asset data.
- REST calls to the instance use OAuth or basic authentication with an integration user. The vendor call uses the vendor's API key, kept in the credential record and not inside the flow.

9. User Interface

Add screenshots of the main screens below.



Add screenshot
Manager approval email



Add screenshot
Turnaround and savings dashboard

10. Testing

- **Strategy:** functional testing of every flow step, performance testing for success rate, execution time and load, and user acceptance testing with the employee, manager and coordinator roles.
- **Tools:** Flow Designer execution details, ServiceNow Automated Test Framework (ATF), a test instance and the UAT template.

11. Screenshots or Demo

Add the demo link and screenshots of the Flow Designer flow and a completed request.



Add screenshot
Completed request with asset record

Demo link: [add demo link here]

12. Known Issues

- The approve button is difficult to tap on small phone screens (BG-003, in progress).
- Vendor lead times are not tracked yet, so delivery dates come only from the vendor's response.

13. Future Enhancements

- Apply the same flow to other standard hardware such as monitors, phones and accessories.
- Compare vendor prices automatically before a purchase order is created.
- Add a mobile-friendly approval screen and chatbot-based requests.
- Predict stock needs from hiring plans and refresh cycles.

Note: the team roles, endpoints, example response and known issues are sample entries. Replace them with your project's real details before you submit.